Bhartiya Vidya Bhavan's Sardar Patel Institute of Technology

DATA ANALYTICS

Title: A Hybrid Classical–Autoencoder–Quantum Framework for Large-Scale Aerosol Regime Discovery Over Asia

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MOTIVATION - PROBLEM STATEMENT

Current Problem

Aerosols strongly affect

1. climate
2. Air quality
3. visibility
4. Human health

→

Their Impacts Depend on

1. Type
2. Size
3. Absorption
4. optical properties

Importance of Classification

Accurate aerosol classification is essential for:

- Climate modelling
- Satellite retrieval improvement
- Air-quality forecasting

Asia contains highly diverse aerosol sources



DATASET - OVERVIEW

A. Source & Geographic Coverage

- Dataset from AERONET (global aerosol network)
- 41 monitoring sites across Asia
- Environments: urban, desert, semi-arid, coastal, island, high-altitude
- Captures major aerosol types: dust, pollution, marine, biomass burning
- Ensures high variability for clustering

B. Parameters & Measurements

- Derived from spectral sun-photometer observations
- 15 key aerosol parameters selected
- Extinction AOD at 440/675/870/1020 nm
- Fine-mode & coarse-mode AOD
- SSA, Absorption AOD, AAE, EAE
- Coincident AOD (440 nm), FMF
- Summary:
- Fine > coarse AOD
- High SSA → scattering aerosols
- Frequency/CDF plots show skewness & variability

C. Data Volume & Preprocessing

- Initial data: 186,562 rows, 48 features
- Removed: missing/invalid/infinite values
- Outlier filtering using robust statistics
- Feature reduction → 15 physically meaningful variables
- Final dataset: 138,262 clean samples

D. Seasonal Segmentation

- Four seasons: DJF | MAM | JJA | SON
- Seasonal influence due to meteorology & emissions
- Observed patterns: MAM: coarse-mode AOD ↑ (dust events), JJA: fine-mode AOD ↑ (pollution, biomass)
- Seasonal stats help interpret temporal aerosol behavior

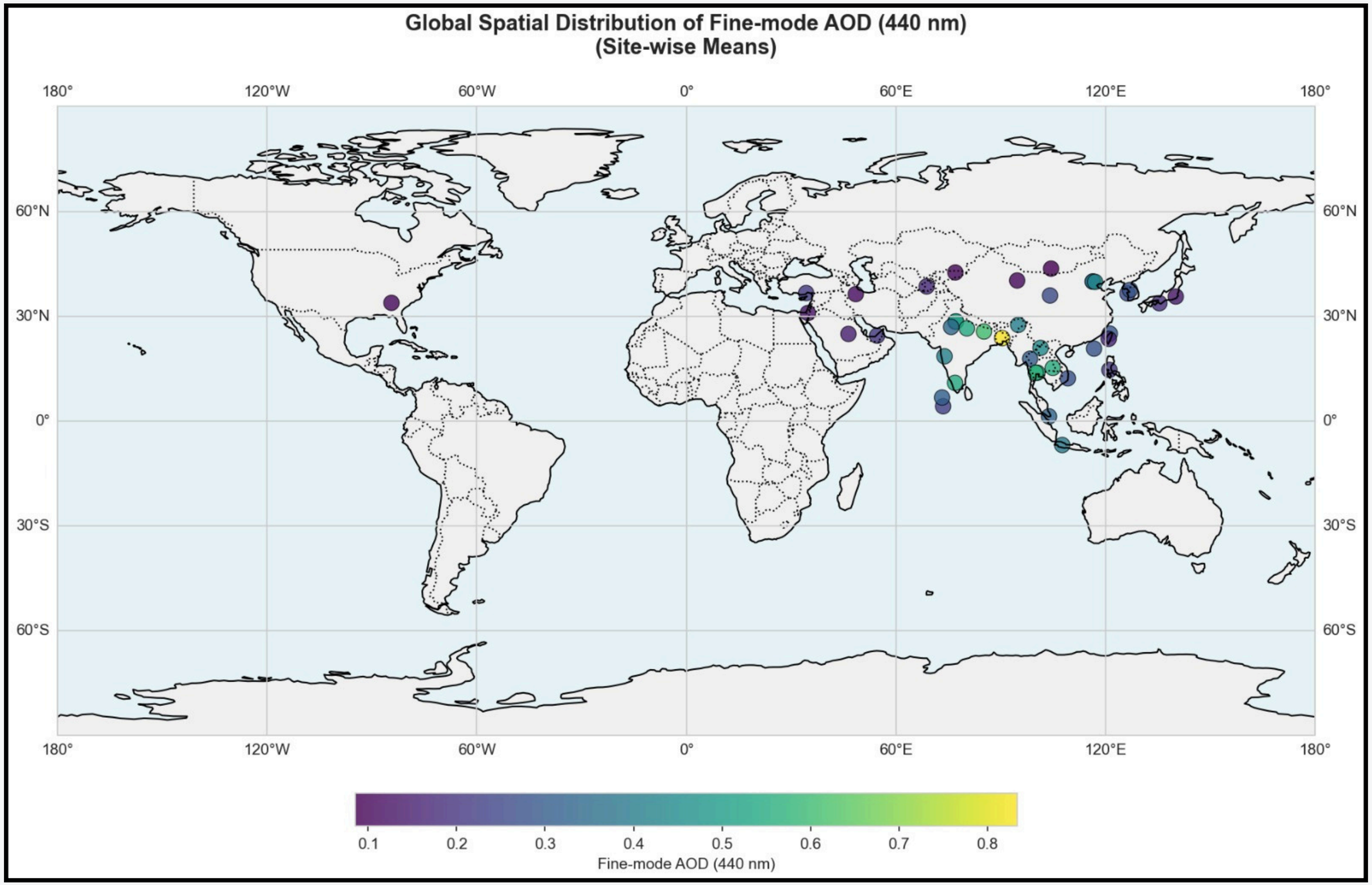


Fig 1 . Frequency distribution and cumulative frequency plots for key aerosol optical parameters.

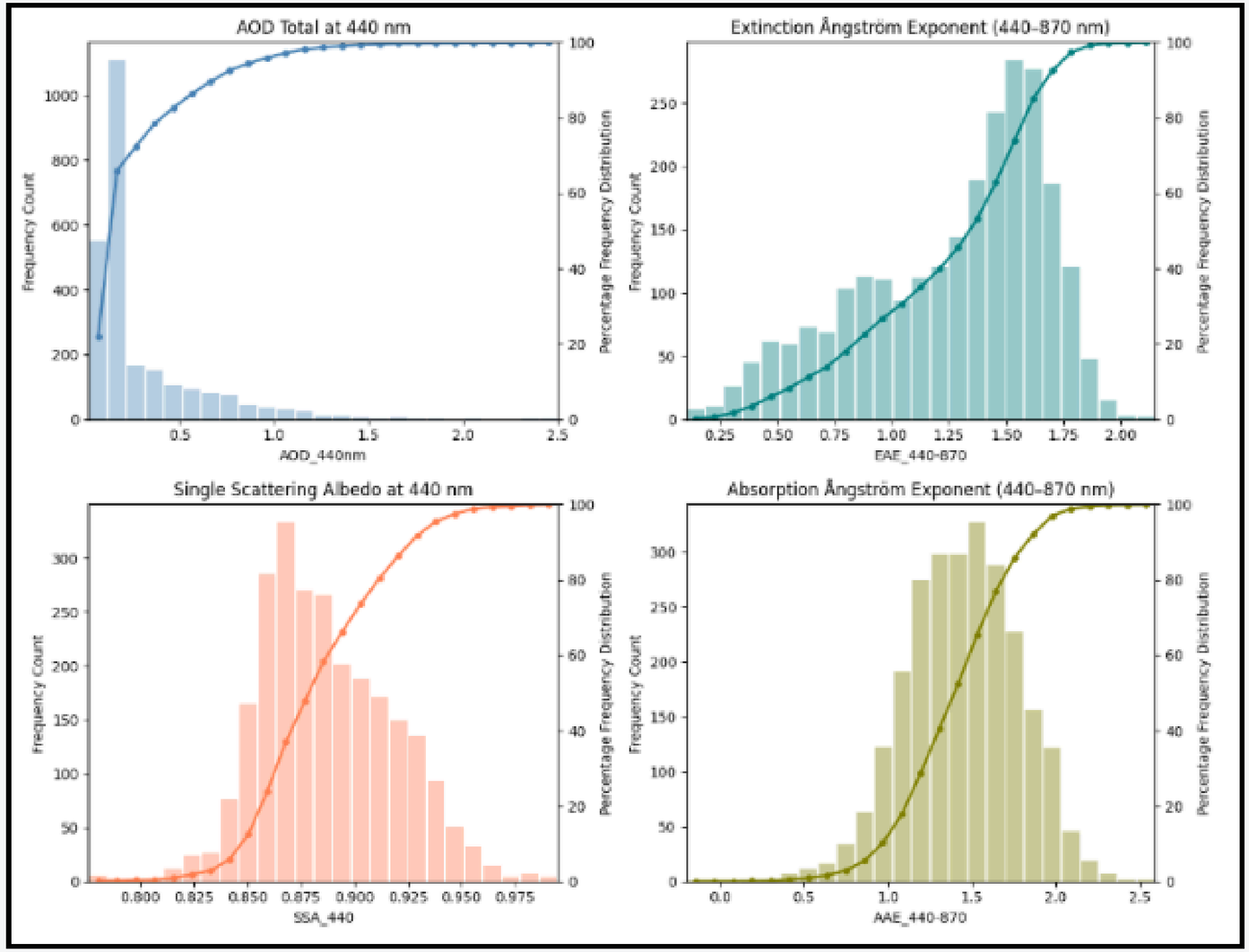


Fig. 1. Frequency distribution and cumulative frequency plots for key aerosol optical parameters, including total AOD at 440 nm, Extinction Ångström Exponent (440–870 nm), Single Scattering Albedo (440 nm), and Absorption Ångström Exponent (440–870 nm). These distributions illustrate the skewness, variability, and spectral behavior of aerosols across the Asian region.

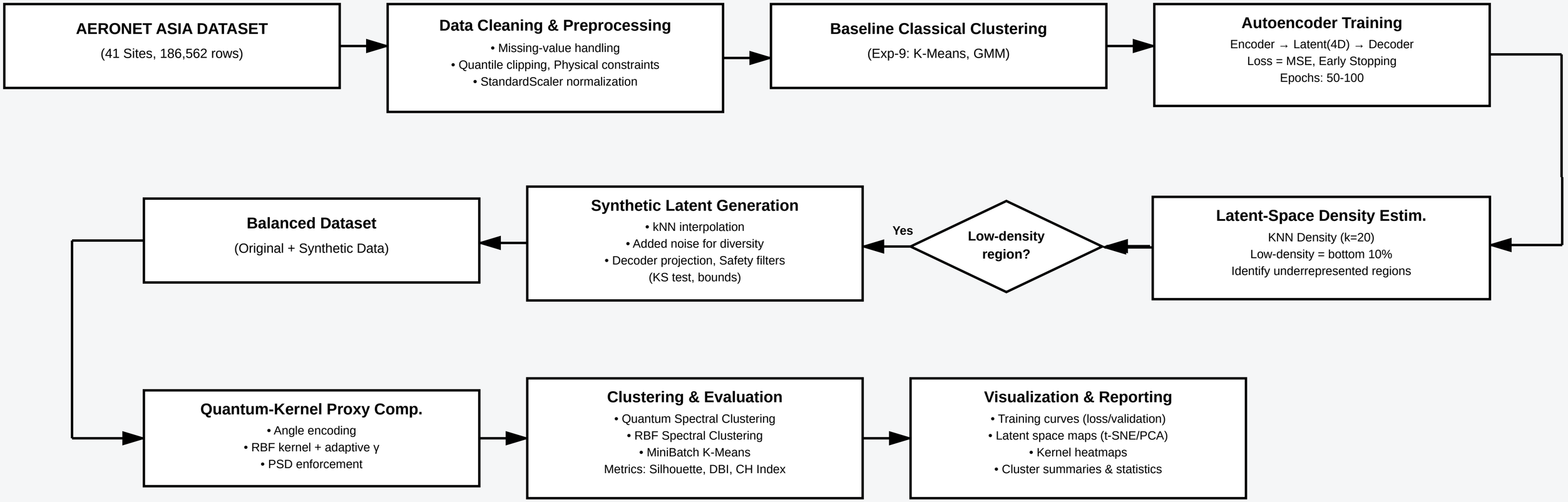
TABLE I: Parameter-wise summary statistics

Parameter	Average ± SD	Median
AOD_T	0.44 ± 0.40	0.32
AOD_F	0.34 ± 0.38	0.20
AOD_C	0.10 ± 0.12	0.06
EAE	1.05 ± 0.43	1.13
SSA	0.89 ± 0.04	0.88
Ref_R	1.48 ± 0.05	1.48
AAOD	0.07 ± 0.03	0.06
AAE	1.50 ± 0.48	1.47

TABLE II: Seasonal aerosol statistics (DJF, MAM, JJA, SON).

Season	AAE	AAOD	AOD_Coarse	AOD_Fine	EAE	SSA
DJF	1.42 ± 0.40	0.07 ± 0.03	0.05 ± 0.06	0.38 ± 0.41	1.19 ± 0.34	0.89 ± 0.03
MAM	1.56 ± 0.53	0.07 ± 0.03	0.15 ± 0.15	0.35 ± 0.36	0.94 ± 0.47	0.89 ± 0.04
JJA	1.55 ± 0.54	0.06 ± 0.03	0.13 ± 0.15	0.28 ± 0.33	0.96 ± 0.47	0.89 ± 0.04
SON	1.48 ± 0.41	0.07 ± 0.03	0.08 ± 0.08	0.32 ± 0.38	1.10 ± 0.37	0.89 ± 0.03

METHODOLOGY



A. Classical Clustering

- **Objective**: Establish a baseline aerosol classification using classical unsupervised methods.
- **Exploratory Insights**

1. AOD distribution is right-skewed
2. EAE indicates fine-mode dominance
3. Moderate absorptivity trends in AAE
4. Observed patterns align with typical Asian aerosol characteristic

- **Clustering Methods Applied**

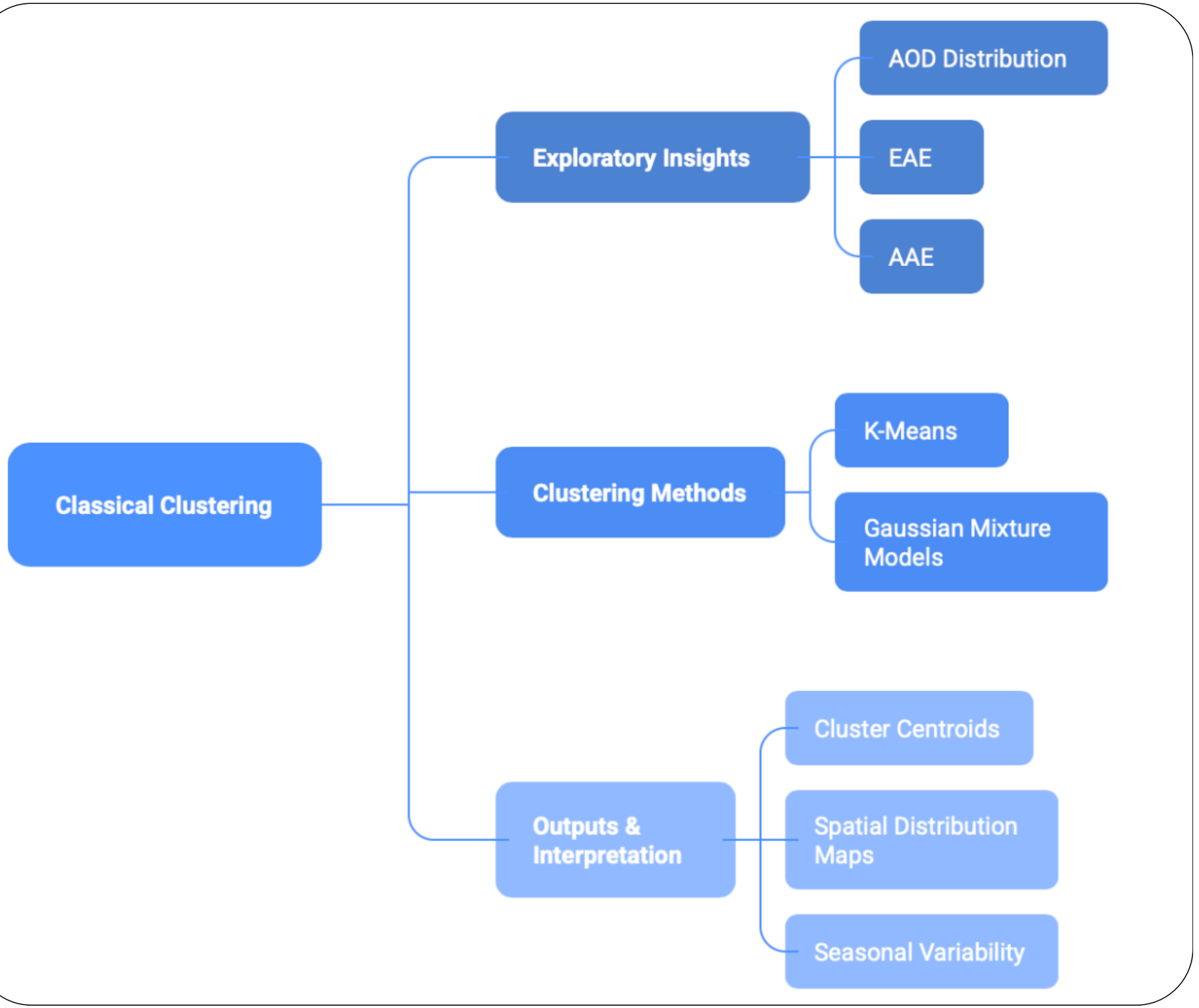
1. K-Means
2. Gaussian Mixture Models (GMM)

Both used to identify 3 dominant aerosol regimes: Dust-dominated, Anthropogenic fine-mode, Mixed aerosols

- **Outputs & Interpretation**

1. Cluster centroids
2. Spatial distribution maps
3. Seasonal variability

- These results form the baseline for later latent-space modeling and augmentation stages.



B. Autoencoder Clustering

- **Objective**: Build a compact latent manifold using a deep autoencoder for density estimation and minority aerosol enhancement.
- **Autoencoder Architecture**

1. Symmetric deep autoencoder
2. Encoder layer sizes: [256 → 128 → 64] → latent dim = 4
3. Decoder mirrors the encoder
4. Training setup:(i) Optimizer: Adam (lr = 3×10^{-4}) (ii) Loss: MSE (iii)Batch size: 512 (iv) Early stopping to avoid overfitting

- **Latent Embedding Generation**

1. Encoder transforms all samples into $Z \in \mathbb{R}^{n\times4}$
2. Latent space analyzed using: (i)PCA (ii) t-SNE (iii) UMAP (iii) Structure visualization highlights aerosol sub-groups

- **Density Estimation**
- k-NN density metric (k = 20)
- Points below the 10th percentile flagged as minority aerosol structures
- Typically represent rare transport events or chemically distinct regimes

- **Generative Oversampling**

Controlled synthesis of new latent samples via:

1. Interpolation
2. Gaussian perturbation

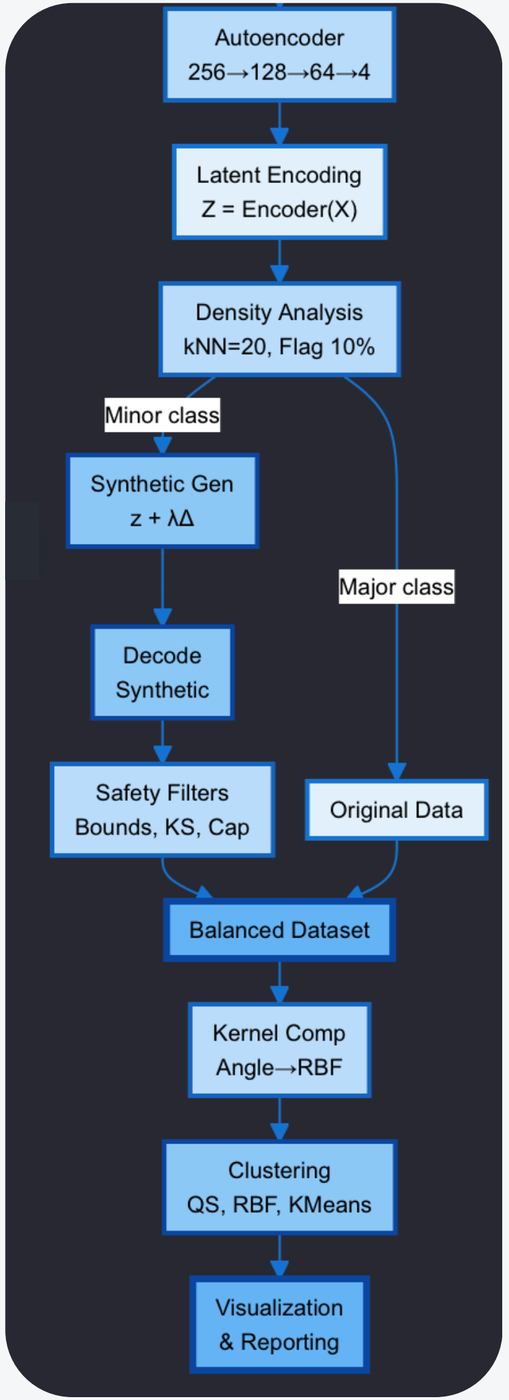
Decoded back to physical aerosol parameters

- **Quality & Physical Validity Checks**

1. Physical constraints applied (SSA, AOD, AE bounds)
2. Distribution consistency: Kolmogorov–Smirnov tests
3. Synthetic volume cap: ≤ 2%
4. Ensures realistic and scientifically valid augmentation

- **Outcome**

1. Balanced dataset with enhanced minority aerosol patterns
2. Supports improved clustering and kernel-based analyses
3. Demonstrated performance gains (ISE-3):
   1. **Silhouette ↑ +60.7%**
   2. **Davies–Bouldin ↓ −42%**





Latent Space

Transform autoencoder output into a 4D latent space.

Angle Encoding

Map each latent dimension to a rotation angle.

Quantum Kernel

Apply RBF kernel on angle-encoded features.

Spectral Embedding

Compute eigenvectors from the graph Laplacian.

Cluster Assignment

Assign data points to clusters using k-Means.

Quantum Clusters

Achieve better separability and stronger boundaries.

C. Quantum Spectral Clustering

- **Objective**: Evaluate whether quantum-inspired similarity measures improve aerosol cluster separability beyond classical methods.
- **Quantum-Proxy Kernel Construction**

1. Use quantum-inspired feature mapping without quantum hardware
2. Angle encoding:

(i)Each latent dimension normalized to a rotation angle in [0, 2π]
(ii)Mimics data-to-qubit mapping in shallow quantum circuits

3.Apply RBF kernel to angle-encoded vectors

4.Hyperparameter tuned via median pairwise distance heuristic

5.Kernel matrix stabilized by:

(i)Eigenvalue clipping
(ii)Optional normalization for contrast

- **Clustering Framework**

1.Quantum-proxy kernel used as affinity matrix
2.Compared against two baselines:
(i)Classical RBF Spectral Clustering
(ii)MiniBatch K-Means
3.Evaluated for k = 2, 3, 4 clusters

- **Evaluation Metrics**

1. Silhouette Score (primary)
2. Davies–Bouldin Index
3. Calinski–Harabasz Score
4. Large reproducible subsample (80k samples) for fair comparison
5. Full-dataset kernels computed with chunking for memory efficiency

- **Outcome**

-->Quantum-proxy Spectral Clustering shows consistently superior separability
-->Achieves highest Silhouette scores across all k-values
-->Indicates that quantum-inspired nonlinear mappings effectively capture complex aerosol structures

RESULTS

A. Initial Clustering Performance

- Spectral Clustering showed the most stable performance
- Optimal cluster count: k = 3
- Silhouette Score: 0.2761 (weak but meaningful separation)
- **Cluster Distribution:**

1. Mixed Aerosols: 73.9%
2. Biomass Burning: 26.1%

- **Cluster Centroids (Key Parameters):**

Mixed Aerosols:
  - AODfine = 0.18
  - EAE = 0.97
  - AAE = 1.56
  - SSA = 0.88

Biomass Burning:
  - AODfine = 0.66
  - EAE = 1.33
  - AAE = 1.25
  - SSA = 0.92

- Patterns match known Asian aerosol regimes (higher fine-mode loading in biomass burning)

TABLE III: Cluster-wise Aerosol Characteristics

Cluster	Percentage	Aerosol Type	Fine AOD	EAE	AAE	RRI
1	73.9	M	0.18	0.97	1.56	1.48
2	26.1	B	0.66	1.33	1.25	1.48

B. Spatial and Seasonal Variability

- **Spatial Patterns**

1. Mixed aerosols dominate most of Asia
2. Biomass Burning aerosols concentrated in: Southeast Asia, Parts of South Asia

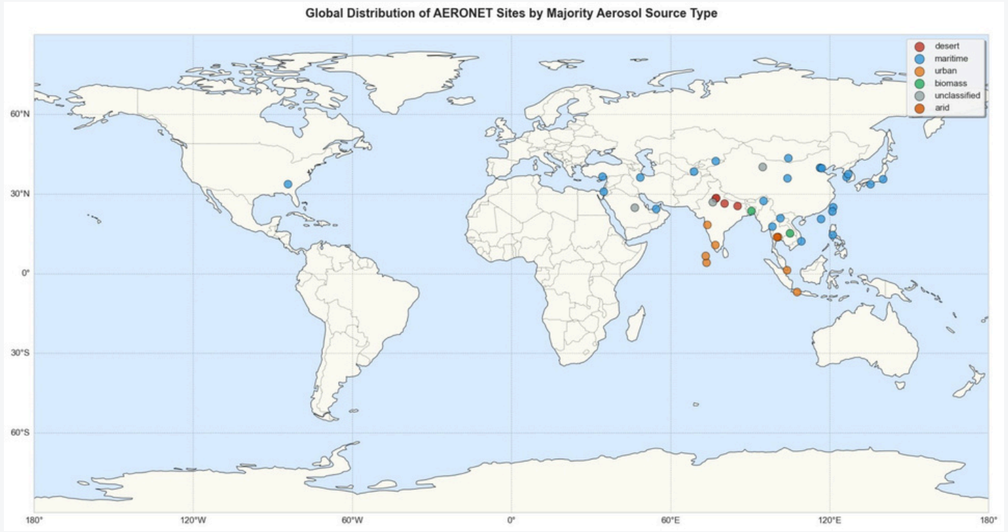
- **Seasonal Cluster Distribution**

DJF: Mixed = 74%
Monsoon: Mixed = 65%
Pre-Monsoon: Mixed = 74%
Post-Monsoon: Mixed = 86%
Remaining 14–35% = Biomass Burning aerosols

- **Seasonal Aerosol Characteristics**

--->Fine-mode AOD highest in Monsoon: **0.35 ± 0.33**

--->Coarse-mode AOD peaks in Pre-Monsoon: **Indicates strong dust activity**



C. Latent-Space Balancing and Synthetic Augmentation

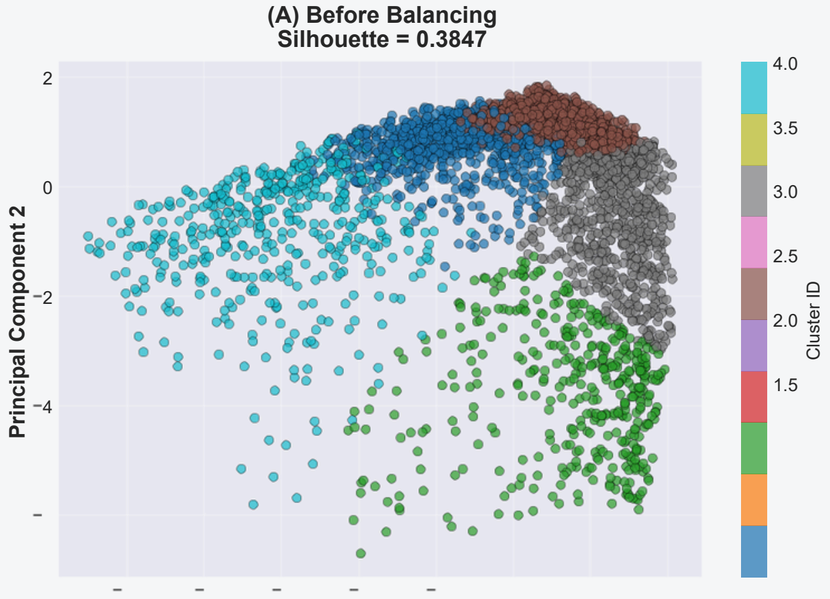
- Autoencoder-based latent sampling used to address class imbalance and strengthen cluster structure
- Autoencoder training performance:

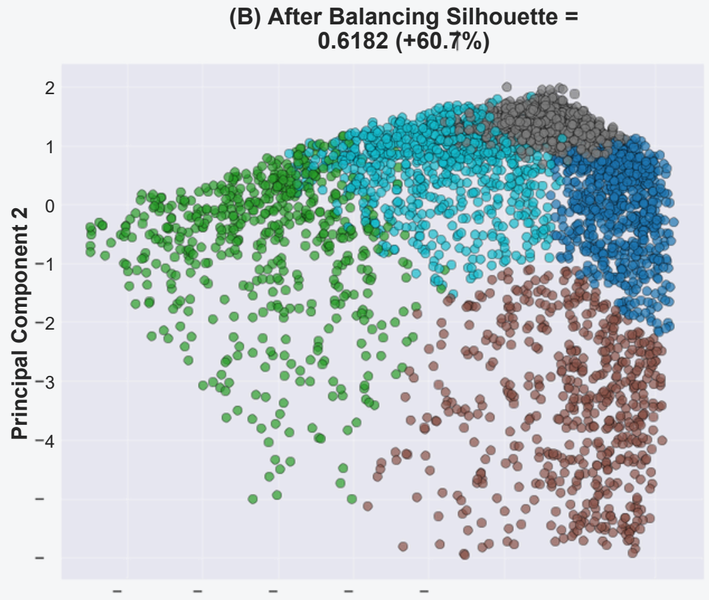
(i) Trained for 22 epochs
(ii) Final MSE = 0.000543
(iii) Validation loss = 0.000621

- Latent dimension = 8, with 37,312 low-density samples flagged for augmentation
- Synthetic generation impact:

(i) Silhouette improved from 0.3847 → 0.6182 (+60.7%)
(ii) Davies–Bouldin index improved from 1.3521 → 0.7843 (42% reduction)
(iii) Calinski–Harabasz increased from 68,423.5 → 124,856.8 (+82.4%)

- PCA visualizations showed synthetic points filling sparse latent regions without distorting manifold
- Feature-wise comparisons confirmed physical validity of augmented samples





D. Kernel Evaluation and Quantum–Classical Comparison

• Classical RBF kernels and quantum kernels evaluated on latent representationsc• Quantum kernel showed strongest cluster separation for all tested k-values

• **Best Quantum Silhouette Score:**

(i) 0.7405 at k = 2 (highest overall)

• **Classical method performance:**

(i) RBF Spectral Clustering:

- 0.2548 (k=2)
- 0.2803 (k=3)
- 0.2959 (k=4)

(ii) K-Means on latent space:

- 0.4189 (k=2)
- 0.4801 (k=3)

• **Quantum Spectral Clustering silhouettes:**

(i) 0.7405 (k=2)

(ii) 0.6263 (k=3)

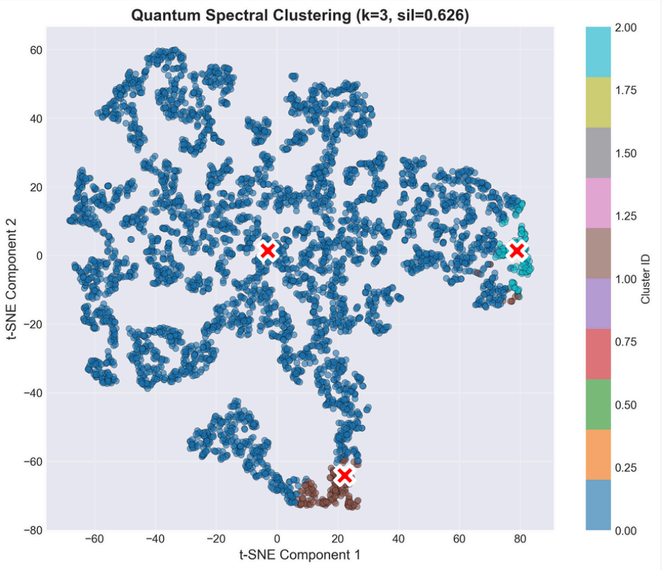
(iii) 0.6211 (k=4)

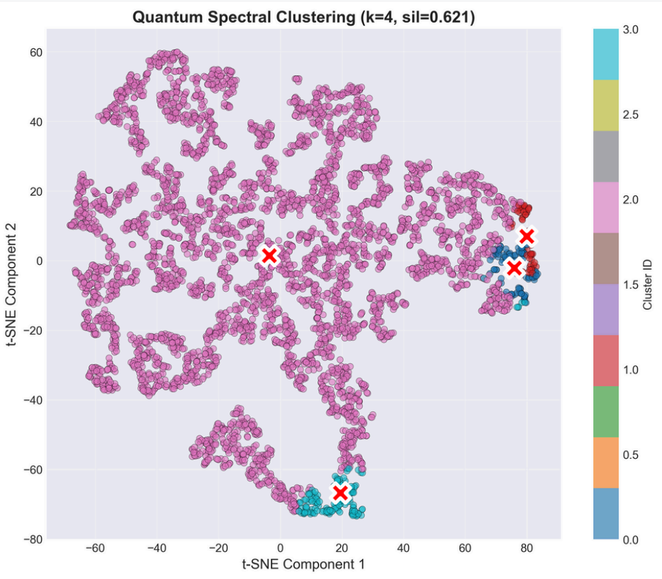
• **Kernel heatmaps showed:**

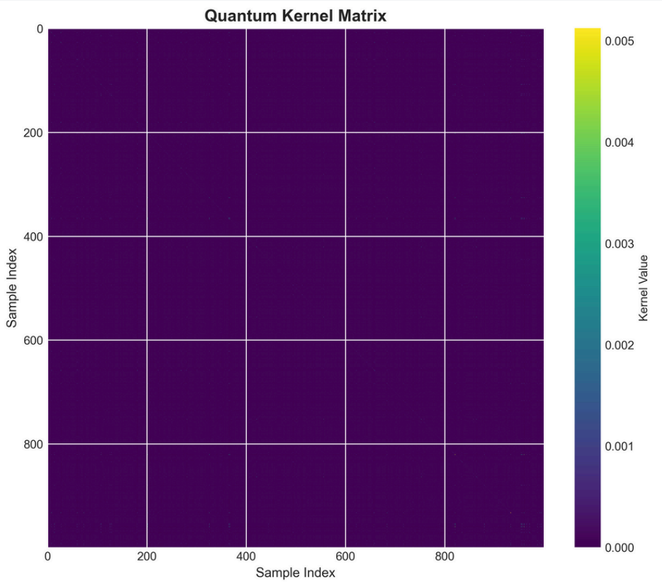
(i) Clearer block structure

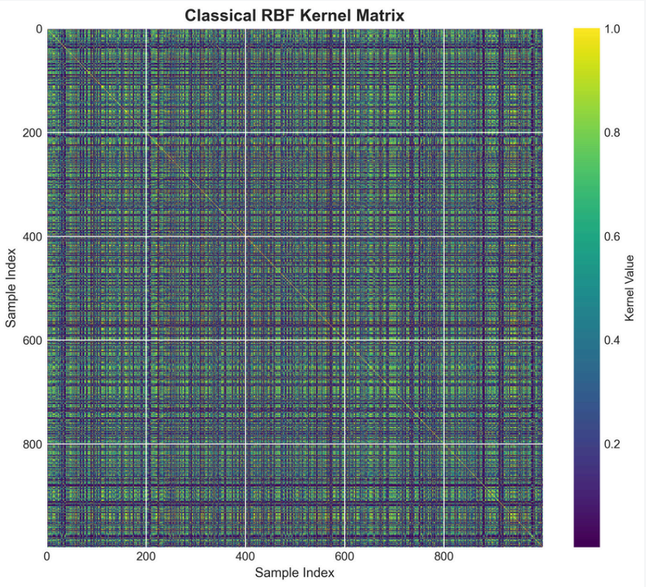
(ii) Stronger diagonal dominance

• **Conclusion**: Quantum kernels capture subtle nonlinear similarities more effectively than classical RBF kernels









CONCLUSION

- Pipeline provides a reproducible and robust framework for large-scale aerosol analysis
- Integrates:

(i) Deterministic preprocessing
(ii) Compact autoencoder for latent-space construction
(iii) Classical + quantum-inspired kernel clustering methods

- Emphasis on stability and interpretability through:

(i) Adaptive kernel tuning
(ii) Density estimation
(iii) Extensive diagnostics

- Quantum component used as a classical proxy, enabling:

(i) Practical evaluation of quantum-inspired similarity measures
(ii) Fair comparison against traditional RBF kernels

- Workflow improves:

(i) Feature structure
(ii) Cluster separability
(iii) Reproducibility across all experiments

- Well suited for:

(i) Future hyperparameter exploration
(ii) Integration of real quantum kernels as hardware matures

THANK YOU !